Somatic copy number alterations are associated with EGFR amplification and shortened survival in patients with primary glioblastoma.

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Glioblastoma (GBM) is the most common malignant primary tumor of the central nervous system. With no effective therapy, the prognosis for patients is terrible poor. It is highly heterogeneous and EGFR amplification is its most frequent molecular alteration. In this light, we aimed to examine the genetic heterogeneity of GBM and to correlate it with the clinical characteristics of the patients. For that purpose, we analyzed the status of EGFR and the somatic copy number alterations (CNAs) of a set of tumor suppressor genes and oncogenes. Thus, we found GBMs with high level of EGFR amplification, low level and with no EGFR amplification. Highly amplified tumors showed histological features of aggressiveness. Interestingly, accumulation of CNAs, as a measure of tumor mutational burden, was frequent and significantly associated to shortened survival. EGFR-amplified GBMs displayed both a higher number of concrete CNAs and a higher global tumor mutational burden than their no EGFR-amplified counterparts. In addition to genetic changes previously described in GBM, we found PARK2 and LARGE1 CNAs associated to EGFR amplification. The set of genes analyzed allowed us to explore relevant signaling pathways on GBM. Both PARK2 and LARGE1 are related to receptor tyrosine kinase/PI3K/PTEN/AKT/mTOR-signaling pathway. Finally, we found an association between the molecular pathways altered, EGFR amplification and a poor outcome. Our results underline the potential interest of categorizing GBM according to their EGFR amplification level and the usefulness of assessing the tumor mutational burden. These approaches would open new knowledge possibilities related to GBM biology and therapy.

Concurrent EGFR amplification and TP-53 mutation in glioblastomas.

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Glioblastoma multiforme is the most common and most aggressive of the primary brain tumors. The mean survival of patients is 10-12 months. Conventional therapy of surgery, radiation and chemotherapy is largely palliative. Cytogenetically, karyotypes of glioblastomas are very complex with trisomy 7 and monosomy 10 as the most frequent abnormalities. A genetic alteration that is significantly more frequent in primary than in secondary glioblastomas, the latter arising from preceding low-grade gliomas, is epidermal growth factor receptor gene (EGFR) amplification, whereas TP-53 mutations are significantly more frequent in low-grade gliomas and secondary glioblastomas derived there- from. We report the histological and genetic study of two glioblastomas, one case arising de novo and the other case arising 3 years after a previously diagnosed anaplastic astrocytoma, with concurrent EGFR amplification and TP-53 mutation. These anomalies were initially deemed as mutually exclusive. However, a small percentage of cases have been found with both anomalies although at a significantly lower level than could be expected. We have analyzed these two cases cytogenetically and by molecular studies in order to detect additional alterations associated with this phenotype. Cytogenetically, both cases showed in common the monosomy of chromosomes 10 and 17. At the molecular level, a rare mutation of TP-53 was found in the secondary glioblastoma and hypermethylation of the promoter region of p16(INK4a) and p14(ARF) genes were observed in the primary and secondary glioblastoma, respectively.

Primary and secondary glioblastomas: from concept to clinical diagnosis.

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Glioblastomas may develop de novo (primary glioblastomas) or through progression from low-grade or anaplastic astrocytomas, (secondary glioblastomas). These subtypes of glioblastoma constitute distinct disease entities that evolve through different genetic pathways, affect patients at different ages, and are likely to differ in prognosis and response to therapy. Primary glioblastomas develop in older patients and typically show EGFR overexpression, PTEN (MMAC1) mutations, CDKN2A (p16) deletions, and less frequently, MDM2 amplification. Secondary glioblastomas develop in younger patients and often contain TP53 mutations as the earliest detectable alteration. These characteristics are derived largely from patients selected on the basis of clinical history and sequential biopsies. Currently available data are insufficient for a substitution of histologic classification and grading of astrocytic tumors by genetic typing alone. More subtypes of glioblastomas may exist with intermediate clinical and genetic profiles, a factor exemplified by the giant-cell glioblastoma that clinically and genetically occupies a hybrid position between primary (de novo) and secondary glioblastomas. Future research should aim at the identification of criteria for a combined clinical, histologic, and genetic classification of astrocytic tumors.

The protein tyrosine phosphatase TCPTP suppresses the tumorigenicity of glioblastoma cells expressing a mutant epidermal growth factor receptor.

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Glioblastoma multiforme (GBM) is the most aggressive type of glioma and GBMs frequently contain amplifications or mutations of the EGFR gene. The most common mutation results in a truncated receptor tyrosine kinase known as Delta EGFR that signals constitutively and promotes GBM growth. Here, we report that the 45-kDa variant of the protein tyrosine phosphatase TCPTP (TC45) can recognize Delta EGFR as a cellular substrate. TC45 dephosphorylated Delta EGFR in U87MG glioblastoma cells and inhibited mitogen-activated protein kinase ERK2 and phosphatidylinositol 3-kinase signaling. In contrast, the substrate-trapping TC45-D182A mutant, which is capable of forming stable complexes with TC45 substrates, suppressed the activation of ERK2 but not phosphatidylinositol 3-kinase. TC45 inhibited the proliferation and anchorage-independent growth of Delta EGFR cells but TC45-D182A only inhibited cellular proliferation. Notably, neither TC45 nor TC45-D182A inhibited the proliferation of U87MG cells that did not express Delta EGFR. Delta EGFR activity was necessary for the activation of ERK2, and pharmacological inhibition of ERK2 inhibited the proliferation of Delta EGFR-expressing U87MG cells. Expression of either TC45 or TC45-D182A also suppressed the growth of Delta EGFR-expressing U87MG cells in vivo and prolonged the survival of mice implanted intracerebrally with these tumor cells. These results indicate that TC45 can inhibit the Delta EGFR-mediated activation of ERK2 and suppress the tumorigenicity of Delta EGFR-expressing glioblastoma cells in vivo.

Glioblastoma-related gene mutations and over-expression of functional epidermal growth factor receptors in SKMG-3 glioma cells.

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Amplification of the epidermal growth factor receptor (EGFR) gene is found in about 40% of glioblastomas

(GBMs) but is rarely detected in GBM cell lines. We confirmed that the exceptional SKMG-3 GBM cell line retained amplified EGFR genes in vitro, and found that these sequences were concentrated on extra-chromosomal DNA particles similar to double-minute chromosomes. The cells contained two other gene mutations that are associated with high-grade astrocytic tumors: extra-chromosomal amplification of the cyclin-dependent kinase-4 (CDK4) gene and a homozygous mutation within the PTEN tumor suppressor gene. Immunoblots revealed very high levels of EGFR, moderately increased expression of CDK4, and no detectable PTEN protein. The overexpressed SKMG-3 EGFRs responded to exogenous ligand and resembled normal rather than mutant receptors. A heterozygous mutation of the p53 gene (p53R282W) correlated with failure of radiation to induce the expression of cyclin-dependent kinase inhibitor p21waf1 or an early G1 cell cycle arrest. Although each of these gene mutations occurs in GBMs, SKMG-3 cells had an unusual genotype in that a p53 gene mutation co-existed with amplified EGFR genes. Nonetheless, the SKMG-3 cell line can be exploited as a model to study how oncogenic EGFR signals in GBM cells interact with over-expressed CDK4 and loss of PTEN to confer the malignant phenotype.

p53 mutation and epidermal growth factor receptor overexpression in glioblastoma.

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Recent molecular studies indicate two different genetic pathways leading to the development of glioblastoma; final progression of astrocytoma and de novo formation. To define the mutual relationships of cytogenetic changes in the pathogenesis of glioblastoma, molecular histopathologic alterations of p53 and epidermal growth factor receptor (EGFR) were evaluated by single stranded conformational polymorphion, reverse transcriptase-polymerase chain reaction and immunohistochemical stains in 15 primary and 21 secondary glioblastomas. Mutations in p53 gene and positive immunoreactivity to p53 protein (DO1) were more prevalent in secondary glioblastomas than in primary glioblastomas. A correlation between p53 mutations and p53 immunopositivities in glioblastomas was observed in 83.3% of the cases. All cases with positive p53 immunoreactivities showed p53 mutations; however, 13.9% of glioblastomas with p53 immuno-positivities lacked the relevant mutations. EGFR amplifications were detected in 73.3% of primary glioblastomas and 9.5% of secondary glioblastomas ($p<0.001$). The concurrence of p53 mutation and EGFR amplification was revealed in only 2 out of 15 primary glioblastomas and none among the secondary glioblastomas. Immunoreactivities for EGFR were noted in 66.7% of primary glioblastomas and in 9.5% of secondary glioblastomas ($p<0.001$). A correlation between EGFR amplification and EGFR immunopositivity in glioblastomas was observed in 91.7% of the cases. These data indicate that EGFR amplification and p53 mutations are two independent genetic events in the development of glioblastomas.

[Genes implicated in glial tumors].

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Because of the absence of specific marker, the histological classification of gliomas remain controversial. Identifying the genetic alterations involved in gliomas makes it possible to define specific molecular pathway of tumoral progression and to define markers of prognostic and diagnostic relevance. For example, p53 mutations are frequent in low grade astrocytoma, anaplastic astrocytoma and secondary glioblastoma suggesting that it takes place at an early stage of development of astrocytic tumors, whereas inactivation of PTEN arises mainly in glioblastomas and EGFR amplification is preferentially associated with "de novo" glioblastoma. Loss of chromosomes 1p and 19q characterizes oligodendroglial tumors. However the putative tumor suppressor genes located on 1p and 19q and specifically inactivated are not known yet. Emerging technologies, like microarrays and microdissection, will allow to refine molecular data and provide a molecular classification of gliomas mechanism involved in the repair of the respiratory epithelium.

High level EGFR amplification in a newly established glioblastoma cell line 170-MG-BA.

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Glioblastoma multiforme is a highly invasive and incurable primary brain tumor. The most frequent genetic alteration therein is amplification of the epidermal growth factor receptor (EGFR) gene, the target of current clinical trials. However, EGFR amplification is poorly represented in glioblastoma cell lines. From the 30 cultures attempted herein, we were able to establish two glioblastoma permanent cell lines. The remaining cultures showed limited life span and underwent senescence between passage numbers (PN) 8 to 15. Our newly established glioblastoma cell lines, designated 170-MG-BA and 538-MG-BA, both originated between PN 3 and 5 when areas of smaller, more rapidly proliferating cells appeared. Both cell lines showed similar rates of growth, moderate morphological differences, cytoskeletal heterogeneity and multiple chromosome rearrangements. Analysis by molecular cytogenetics and comparative genomic hybridization (aCGH) revealed two copies of a stable marker chromosome in 170-MG-BA cells effecting focal amplification at 7q11 of the EGFR locus. Comparative RqPCR analysis confirmed that EGFR was uniquely highly expressed in 170-MG-BA cells. Combined targeted expression analysis and aCGH data excluded the recurrent EGFRvIII activating mutation. In contrast, EGFR expression in 538-MG-BA cells which lacked genomic EGFR amplification was not raised. Immunofluorescent staining showed high EGFR protein expression only in the 170-MG-BA cells. Cytogenetic, genomic and transcriptional analyses then confirmed high-level genomic amplification and transcriptional upregulation of wild type EGFR in 170-MG-BA; the first conventional cell line model for investigating the biology and targeted therapy of this key alteration in glioblastoma. Both cell lines are freely available from the DSMZ cell repository.

EGFRvIII mutations can emerge as late and heterogenous events in glioblastoma development and promote angiogenesis through Src activation.

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Amplification of the epidermal growth factor receptor (EGFR) and its mutant EGFRvIII are among the most common genetic alterations in glioblastoma (GBM), the most frequent and most aggressive primary brain tumor. In the present work, we analyzed the clonal evolution of these major EGFR aberrations in a small cohort of GBM patients using a unique surgical multisampling technique. Furthermore, we overexpressed both receptors separately and together in 2 patient-derived GBM stem cell lines (GSCs) to analyze their functions in vivo in orthotopic xenograft models. In human GBM biopsies, we identified EGFR amplification as an early event because EGFRvIII mutations emerge from intratumoral heterogeneity later in tumor development. To investigate the biological relevance of this distinct developmental pattern, we established experimental model systems. In these models, EGFR+ tumor cells showed activation of classical downstream signaling pathways upon EGF stimulation and displayed enhanced invasive growth without evidence of angiogenesis in vivo. In contrast, EGFRvIII+ tumors were driven by activation of the prototypical Src family kinase c-Src that promoted VEGF secretion leading to angiogenic tumor growth. The presented work shows that sequential EGFR amplification and EGFRvIII mutations might represent concerted evolutionary events that drive the aggressive nature of GBM by promoting invasion and angiogenesis via distinct signaling pathways. In particular, c-SRC may be an attractive therapeutic target for tumors harboring EGFRvIII as we identified this protein specifically mediating angiogenic tumor growth downstream of EGFRvIII.

Genetic markers in glioblastoma: prognostic significance and future therapeutic implications.

PMID: 12826827 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the highest-grade infiltrative astrocytoma and also the most common. It is generally associated with a dismal prognosis (mean survival 11 months), yet individual patient survivals vary. Histologic parameters have had limited value in predicting survival among patients with GBM. The current view of GBM as a histopathologic entity consisting of several genetic subtypes raises the possibility that molecular alterations could be predictive of survival. Common genetic alterations in GBM include gene amplification of epidermal growth factor receptor (EGFR), mutations in the tumor suppressors TP53 and PTEN, and genetic losses on chromosome 10. Less common in GBMs is the combined loss of chromosomes 1p and 19q-a combination that has proven prognostically favorable in oligodendrogliomas. A recent article on prognostic factors in a series of 97 GBMs by Schmidt et al. finds that both TP53 mutations and young patient age at presentation are independent factors associated with a long survival. Loss of heterozygosity (LOH) of chromosome 10q was predictive of a poor outcome. Perhaps most intriguing, the finding of combined LOH of 1p and 19q, which was noted in only five GBMs, was associated with a significantly longer survival. Thus, combined losses of 1p and 19 may be associated with a favorable prognosis in a wider range of infiltrative gliomas that includes GBM. While these findings will be debated and need to be confirmed, it is clear that genotyping of infiltrative gliomas will be an important component of neuro-oncology in the future. Not only will genetic alterations offer prognostication, but they will also serve as targets for directed therapies. Treatments directed against tumors with EGFR amplification, TP53, mutations and PTEN mutations are being developed and tested in clinical trials. It remains to be seen if GBMs with 1p and 19q losses are chemosensitive in the same manner as oligodendrogliomas.